E-Commerce Sales & Customer Intelligence

Project Report | Python • MySQL • Power BI

1. Executive Summary

This project analyzes the Olist Brazilian E-Commerce Public Dataset to explore order performance, item sales, payment behavior, delivery experience, product categories, customer geography, and review scores. Data was prepared and loaded into MySQL, checked through 17 validation checks, analyzed with SQL, and summarized in a Power BI dashboard.

2. Objectives

- Prepare and validate multiple related e-commerce tables.
- Use SQL to answer business-focused questions about orders, sales, payments, delivery, and reviews.
- Build a dashboard that communicates high-level KPIs and patterns.
- Identify practical areas for further investigation, especially delivery performance and customer satisfaction.

3. Tools and Dataset

Tools: Python/Pandas for data preparation, MySQL 8.0 for storage and SQL analysis, and Power BI Desktop for visualization. The database contains nine tables from the Olist dataset.

Table	Rows
Customers	99,441
Orders	99,441
Order items	112,650
Payments	103,886
Reviews	99,224
Products	32,951
Sellers	3,095
Geolocation	738,332
Category translation	71

The project's database validation reported that all 17 checks passed.

4. Key Findings

Order performance

There were 99,441 orders, of which 96,478 were delivered (approximately 97.02%). The remaining orders were distributed across shipped, canceled, unavailable, invoiced, processing, created, and approved statuses.

Product categories

Category	Delivered item sales (BRL; excluding freight)
Health & Beauty	R$1,233,131.72
Watches & Gifts	R$1,166,176.98
Bed, Bath & Table	R$1,023,434.76
Sports & Leisure	R$954,852.55

Category	Delivered item sales (BRL; excluding freight)
Computers & Accessories	R$888,724.61

Delivery and reviews

Average delivery time was 12.50 days across 96,470 delivered orders with the required dates; the late-delivery rate was 8.11%. Late-delivered orders had an average review score of 2.57, and 54.03% received a score of 1 or 2. Orders delivered on time or early had an average score of 4.29, with 9.23% receiving a score of 1 or 2. This is an association, not proof that delivery timing alone caused the review differences.

Payment behavior

Credit card was the most common payment method by payment records (76,795 records; approximately R$12.54 million in recorded payment value). Boleto followed with 19,784 records and approximately R$2.87 million.

Customer geography and repeat purchasing

São Paulo (SP) had the largest customer count at 40,302. The repeat-customer analysis classified 93,099 as one-time and 2,997 as repeat customers using customer_unique_id and the project's grouping logic.

5. Dashboard Contents

The Power BI dashboard contains KPI cards for total orders, total item sales, average item value, and average payment value, plus visuals for monthly sales trend, payment method, top product categories, review score distribution, customers by state, and monthly order volume.

6. Data and Interpretation Notes

- Item sales based on order_items.price exclude freight and may include orders with statuses other than delivered unless a filter is applied.
- Average payment value and average item price are different measures and should not be compared as if they were the same metric.
- The first and last months in the dataset may be partial periods.
- Confirm Power BI relationships and ensure the sales trend displays monthly values, not a cumulative total, before presenting final conclusions.

7. Recommendations

Investigate late-delivery causes by seller, product category, and destination state; monitor low review scores alongside delivery performance; and use a consistent definition for sales and delivered orders across SQL and Power BI. Further work could include a documented data dictionary, reproducible setup steps, and additional validation of dashboard measures.

8. Reproducibility and Repository

Suggested repository folders are data/, src/, sql/, reports/, and dashboard/, plus this README.md. Reproduction requires the dataset files, Python scripts and dependencies, SQL scripts, a configured MySQL database, and the Power BI file. Do not publish passwords or credentials. Keep large raw files out of GitHub if repository limits require it.

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